

RONAK NARKHEDE

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EDUCATION

University of Minnesota | Master of Science in Robotics

St Paul, MN

GPA: 3.3/4.0; Relevant Coursework: Robotics, Robot Vision, Machine Learning

Sept 2025 – May 2026

SRM University | Bachelor of Technology in Mechatronics

Chennai, India

GPA: 8.32/10.0

Sept 2020 – May 2024

TECHNICAL SKILLS

Programming Languages: C++, Python, MATLAB

Libraries & Frameworks: ROS 1, MoveIt, OpenCV, Eigen, PCL, PyTorch, NumPy, SciPy

Tools & Platforms: Docker, Linux, SolidWorks, Onshape, Git, MuJoCo, Gazebo, Robot Programming (RoboDK, ABB RAPID, URScript)

PROFESSIONAL EXPERIENCE

Medical Robotics and Devices Lab | Graduate Research Assistant Minneapolis, MN | Sept 2025 – Present

- Built a real time motion tracking system using an electromagnetic sensor for 6 DOF pose estimation in surgical navigation, integrating this tracking pipeline with a tendon driven soft robot to form a closed loop control system using a data driven Jacobian estimate, eliminating the need for an analytical kinematic model.
- Developed and deployed a multi threaded, teleoperation platform for remote surgical robot control using FastAPI Webserver, WebSockets, RESTful APIs, and WebRTC streaming.
- Used OpenCV and ArUco marker for guidewire detection and implemented a feature enforcing safety stop to disallow the guidewire from fully retracting from the vascular phantom.
- Defined communication protocols between the server and an ESP32, allowing for low latency control with avg 80 ms response times.
- Partnered with Dr. Andrew Grande and Medtronic to pursue ARPA-H AIR federal grant funding, advancing past initial screening to full proposal stage.

Choice Robotics Lab | Research Volunteer

Minneapolis, MN | Jan 2025 – April 2025

- Engineered a learning framework for the ALOHA bimanual robot using MuJoCo and Gymnasium, focusing on contact rich tasks like Cube Transfer and precision Peg-in-Hole Insertion.
- Developed a hybrid Behavioral Cloning + Soft Actor-Critic pipeline, pre-training the policy on 50 expert demonstrations to control exploration bottlenecks before fine-tuning via reinforcement learning.
- Optimized sample efficiency by 23% and achieved a 78% success rate on transfer cube task.

PROJECTS

UR5 Manipulation & Force Control | UR5, URScript, Python, RoboDK

Fall 2025

- Programmed a UR5 robot for automated flashlight assembly using Python (RoboDK) and URScript, with pneumatic grippers via digital I/O and force controlled threading.
- Implemented Cartesian Impedance Control using raw force sensor feedback for compliant peg-in-hole insertion without any vision feedback.
- Tuned system parameters like TCP offsets and Center of Gravity to improve positioning accuracy and enable reliable force controlled operations.

Vision Based Robot Control | Stereo Camera, ArUco, ROS, C++

Fall 2024

- Built wireless vision based teach pendant for 3-DOF robot arm using stereo camera and ArUco markers for hand-eye calibration and intuitive 3D point selection
- Engineered a ROS control pipeline to transform selected 3D coordinates from the teach pendant into target joint angles using numerical Inverse Kinematics, driving the end effector to the selected location.

PUBLICATIONS

Towards Remote Thrombectomy with Telerobotically-Driven Guidewires	<i>Accepted</i>
R. Narkhede, et al. <i>Design of Medical Devices Conference (DMD)</i>	April 2026
Estimating the Non-parametric Jacobian of a Tendon-driven Soft Robot	<i>Accepted</i>
R. Narkhede, et al. <i>Design of Medical Devices Conference (DMD)</i>	April 2026

LEADERSHIP EXPERIENCE

Next Tech Lab Board Member, Robotics and Embedded Systems	Chennai, India Apr 2022 – June 2024
- Organized over 20+ talks, 5 hackathons, and 3 research seminars. - Recruited & led a team of over 30 undergraduate researchers over 2 years, supervising 20+ projects.	